

# NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

## Sixth Semester B.E. (Mechanical) (Credit System) Degree Examinations

May - June 2023

### 20ME603 - AUTOMOTIVE ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

|                   |                                                                                                         | Marks | BT* | CO* | PO* |
|-------------------|---------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| <b>Unit – I</b>   |                                                                                                         |       |     |     |     |
| 1. a)             | List the components of automotive engine. Mention their functions and materials used for manufacturing. | 6     | L*1 | 1   | 1   |
| b)                | Describe valve timing diagram for a four stroke Otto cycle engine with the help of a diagram.           | 6     | L2  | 1   | 1   |
| c)                | With a neat sketch, explain the working of connecting rod.                                              | 8     | L2  | 1   | 1   |
| 2. a)             | What are the needs of engine lubrication? Explain pressure feed lubrication with neat sketch.           | 10    | L2  | 1   | 1   |
| b)                | Sketch and explain four types of combustion chambers of S.I engines.                                    | 10    | L2  | 1   | 1   |
| <b>Unit – II</b>  |                                                                                                         |       |     |     |     |
| 3. a)             | With a neat sketch explain single point fuel injection system.                                          | 10    | L2  | 2   | 1   |
| b)                | Sketch and explain the working principle of simple carburetor.                                          | 10    | L2  | 2   | 1   |
| 4. a)             | With neat circuit diagram explain the principles of battery ignition system.                            | 8     | L2  | 2   | 1   |
| b)                | Sketch and explain the working principle of centrifugal and vacuum advance ignition mechanism.          | 12    | L2  | 2   | 1   |
| <b>Unit – III</b> |                                                                                                         |       |     |     |     |
| 5. a)             | With a neat sketch explain the working principle of multiplate clutch.                                  | 10    | L2  | 3   | 1   |
| b)                | Sketch and explain the working principle of fluid coupling.                                             | 10    | L2  | 3   | 1   |
| 6. a)             | With a neat sketch explain the working of constant mesh gear box.                                       | 10    | L2  | 3   | 1   |
| b)                | With a neat sketch explain the working principle of centrifugal clutch.                                 | 10    | L2  | 3   | 1   |
| <b>Unit – IV</b>  |                                                                                                         |       |     |     |     |
| 7. a)             | What is function of differential and explain its operation with neat diagram.                           | 12    | L2  | 4   | 1   |
| b)                | Explain the working of Torque Tube drive with a neat sketch.                                            | 8     | L2  | 4   | 1   |
| 8. a)             | Discuss air suspension system with a neat sketch.                                                       | 10    | L2  | 4   | 1   |
| b)                | Sketch and explain the working principle of rack and pinion steering gear system.                       | 10    | L2  | 4   | 1   |
| <b>Unit – V</b>   |                                                                                                         |       |     |     |     |
| 9. a)             | Sketch and explain the working principle of master cylinder.                                            | 10    | L2  | 5   | 1   |
| b)                | With a neat diagram explain the working principle of hydraulic braking system.                          | 10    | L2  | 5   | 1   |
| 10. a)            | What are the desirable properties of tyres?                                                             | 4     | L2  | 5   | 1   |
| b)                | Write short note on emission standards.                                                                 | 6     | L2  | 5   | 1   |
| c)                | With neat sketch, explain exhaust gas recirculation system.                                             | 10    | L2  | 5   | 1   |

BT\* Bloom's Taxonomy, L\* Level, CO\* Course Outcome, PO\* Program Outcome.

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